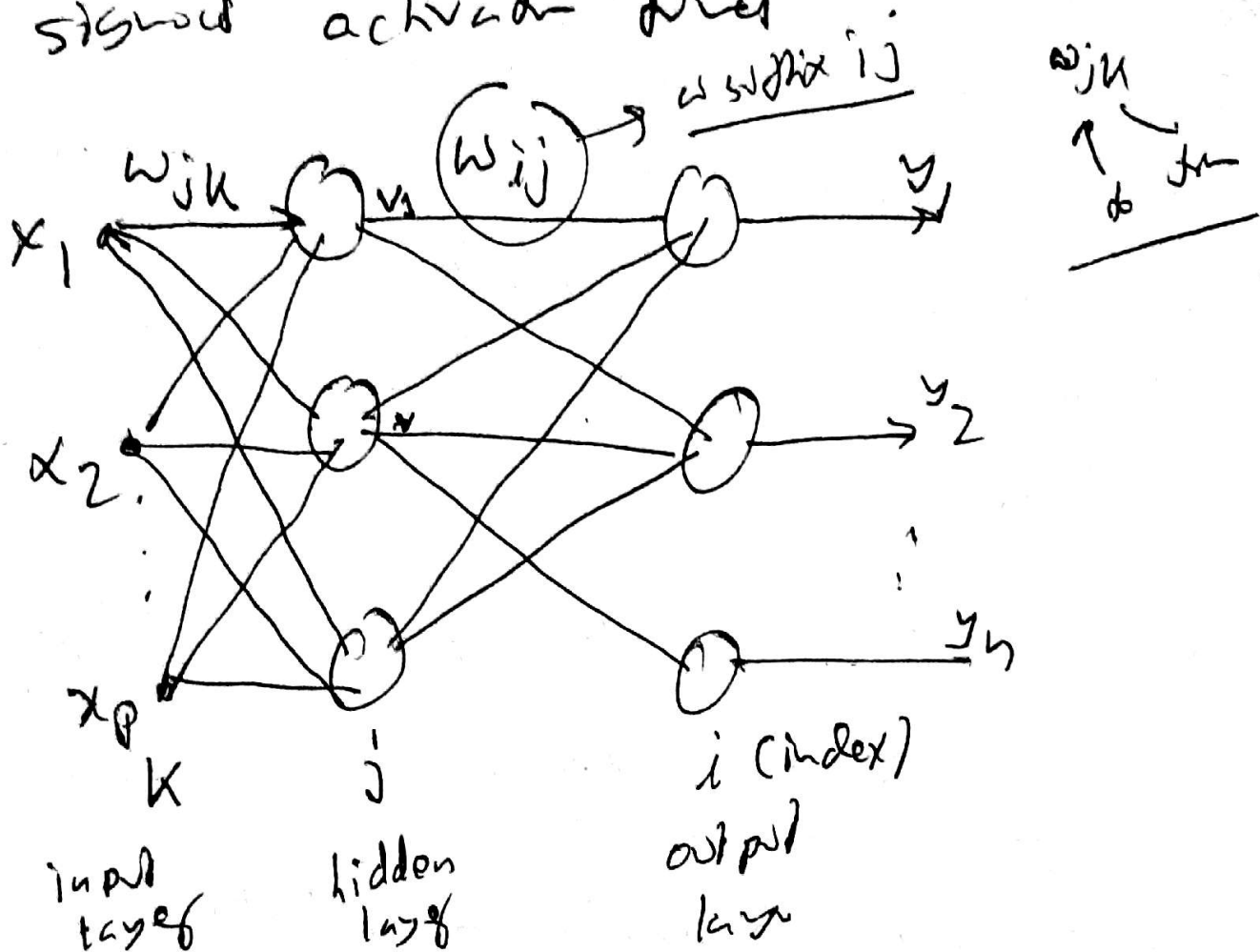


A two layered feed forward nn with sigmoid activation func



$w_{ij} \rightarrow$ is a typical weight connecting j th neuron in the hidden layer to i th neuron in the output layer

$w_{jk} \rightarrow$ is a typical weight connecting k th neuron in the input layer to j th neuron in the hidden layer

The output of hidden unit n is
Sigmoidally activated

$$v_j = \frac{1}{1 + e^{-h_j}}$$

$$h_j = \sum W_{jk} x_k$$

$$y_i = \frac{1}{1 + e^{-s_i}}$$

$$s_i = \sum W_{ji} v_j$$

with neuron

Create the cost function (Error)

$$E = \frac{1}{2} \sum (y_i^d - y_i)^2$$

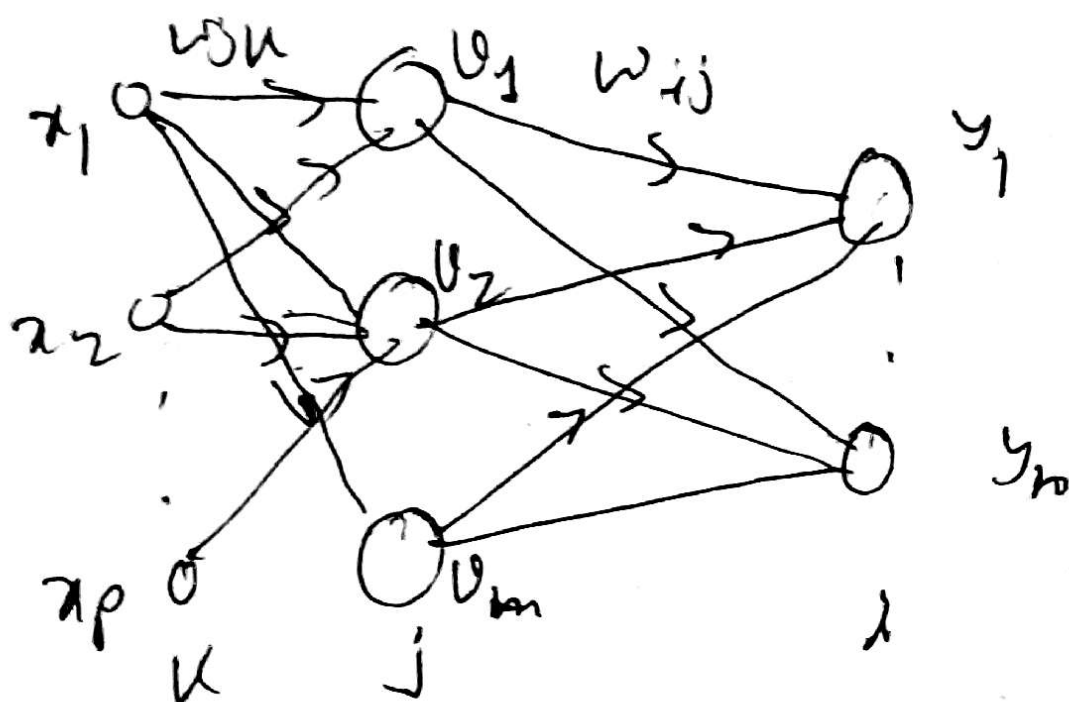
The gradient descent rule
weight update rule for weights betⁿ the
hidden layer & the output layer

$$w_{ij}(t+1) = w_{ij}(t) - \eta \frac{\delta E}{\delta w_{ij}}$$

weight update rule for weights betⁿ input
layer & the hidden layer

$$w_{jk}(t+1) = w_{jk}(t) - \eta \frac{\delta E}{\delta w_{jk}}$$

$\frac{\delta E}{\delta w_{ij}}$ & $\frac{\delta E}{\delta w_{jk}}$ have to be determined



$$v_j = \frac{1}{1 + e^{-h_j}} \quad ; \quad h_j = \sum w_{jk} x_k$$

$$y_i = \frac{1}{1 + e^{-s_i}} \quad , \quad s_i = \sum w_{ij} v_j$$

$$s_i = w_{i1}v_1 + w_{i2}v_2 + \dots + w_{im}v_m$$

$y_1^d, y_2^d, \dots, y_3^d, \dots, y_n^d$ given:

$y_1, y_2, y_3, \dots, y_n$ (computed)

$$E = \frac{1}{2} \sum_i (y_i^d - y_i)^2 \quad [\text{cost function / instantaneous}]$$

$$w_{ij}(t+1) = w_{ij}(t) - \eta \frac{\partial E}{\partial w_{ij}} \quad \text{--- (1)}$$

weight update for weight betⁿ hidden layers and output layer

$$\begin{aligned} \frac{\partial E}{\partial w_{ij}} &= \frac{\partial}{\partial w_{ij}} \left(\frac{1}{2} \sum (y_i^d - y_i)^2 \right) \\ &= \frac{\partial}{\partial w_{ij}} \left(\frac{1}{2} y_i^d - y_i \right)^2 \end{aligned}$$

[Since E is the function of weight & hence E (system output)]

$$\begin{aligned} &= \frac{\partial}{\partial w_{ij}} \frac{1}{2} (y_i^d - y_i)^2 \\ &= -(y_i^d - y_i) \frac{\partial y_i}{\partial w_{ij}} \quad \text{--- (2)} \end{aligned}$$

$$y_i = \frac{1}{1 + e^{-s_i}}$$

$$\frac{\partial y_i}{\partial w_{ij}} = \frac{\partial y_i}{\partial s_i} \cdot \frac{\partial s_i}{\partial w_{ij}}$$

$$y_i = \frac{1}{1 + e^{-s_i}} \quad \frac{d(x)(1 - f(x))}{dx}$$

$$\frac{dy_i}{ds_i} = y_i (1 - y_i) \quad (1)$$

$$\frac{ds_i}{d\omega_{ij}} = \delta \left(\sum \omega_{ij} v_j \right)$$

$$= v_j$$

$$\frac{\delta E}{\delta \omega_{ij}} = -(y_i^d - y_i) \frac{\delta y_i}{\delta \omega_{ij}} \quad (2)$$

$$= -(y_i^d - y_i) y_i (1 - y_i) v_j \quad (3)$$

$$\omega_{ij}(t+1) = \omega_{ij}(t) + \eta (y_i^d - y_i) y_i (1 - y_i) v_j$$

$$\omega_{ij}(t+1) = \omega_{ij}(t) + \eta \delta_i v_j$$

$$\text{where } \delta_i = y_i (1 - y_i) (y_i^d - y_i)$$

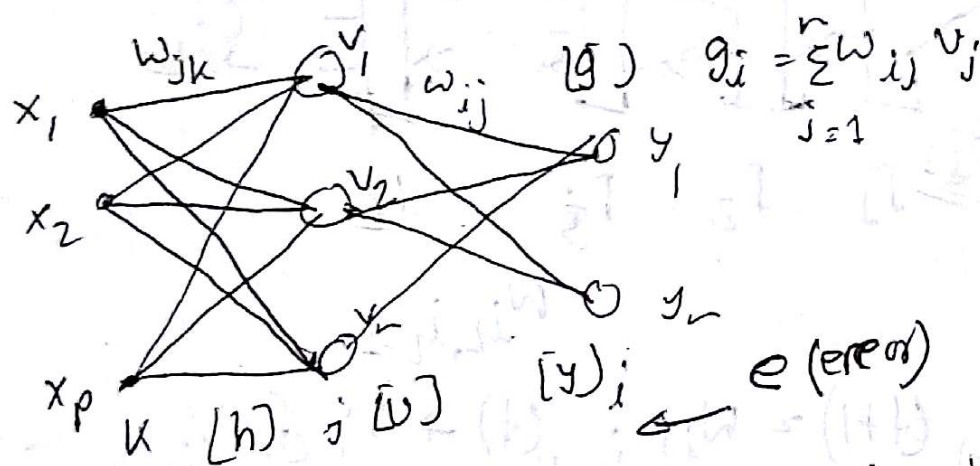


$$w_{jk}(t+1) = w_{jk}(t) + \eta \delta_j x_k$$

$$\delta_j = v_j (1 - v_j) \sum \delta_i w_{ij}$$

(Exercise 20.4.1)

$$\delta_i = y_i (1 - y_i) \delta_i (y_i^d - y_i)$$



$$h_j = \sum_k w_{jk} x_k \quad v_j = \sum_k w_{jk} x_k \quad y = \sum_i w_{ij} v_i$$

$$= \frac{1}{1 + e^{-\sum_i w_{ij} x_k h_j}} = \phi(h_j)$$

$$y_i = \phi(g_i) = \frac{1}{1 + e^{-g_i}}$$

The error function that we seek to minimize

$$E(t) = \frac{1}{2} (y_i^d(t) - y_i(t))^2$$

Where y_i^d is the desired response of the i th unit of the output layer. The update law of the weights

Gradient Descent rule

$$W_{ij}(t+1) = W_{ij}(t) - \eta \frac{\delta E(t)}{\delta W_{ij}(t)}$$

Updating weights connecting output layer and hidden layer

Computing δE

$$\frac{\delta E(t)}{\delta W_{ij}(t)} = \sum \frac{\delta E(t)}{\delta g_i} \times \frac{\delta g_i}{\delta W_{ij}(t)}$$

$$\frac{\delta E(t)}{\delta y(t)} = \frac{1}{2} \times 2 (y_i^d - y_i) \times (-1)$$

$$y_i = \frac{1}{1 + e^{-x}}$$

$$\frac{\delta y_i}{\delta W_{ij}} = \frac{\delta y_i}{\delta g_i} \times \frac{\delta g_i}{\delta W_{ij}}$$

Update law

Since $\frac{\delta y_i}{\delta g_i} = y_i(1 - y_i) \quad \frac{\delta g_i}{\delta W_{ij}} = v_j$

$$\delta - \Delta W_{ij} = \eta (y_i^d - y_i) y_i(1 - y_i) v_j \quad \text{input}$$

Therefore

$$W_{ij}(t+1) = W_{ij}(t) + \eta (y_i^d - y_i) y_i(1 - y_i) v_j$$

$$= W_{ij}(t) + \eta \delta_i v_j$$

$\delta_i = y_i(1 - y_i) (y_i^d - y_i)$ is the error back output layer



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Updating weights connecting hidden layer and input layer

$$\frac{\partial E(t)}{\partial w_{jk}(t)} = \sum \frac{\partial E(t)}{\partial y_j(t)} \times \frac{\partial y_j}{\partial w_{jk}(t)}$$

$$\frac{\partial E}{\partial y_i} = (-1) (y_i^d(t) - y_i(t))$$

$$\frac{\partial y_i}{\partial w_{jk}} = \frac{\partial y_i}{\partial v_j} \times \frac{\partial v_j}{\partial w_{jk}}$$

Update law

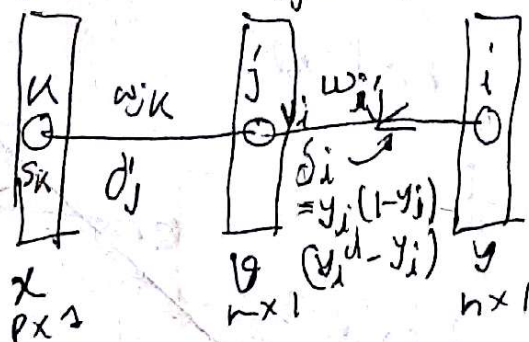
$$w_{jk}(t+1) = w_{jk}(t) + \eta \delta_j x_k$$

$$\delta_j = v_j (1 - v_j) \sum_{i=1}^n \delta_i w_{ij}$$

Back propagation algorithm

0 < \eta < 1

$$w_{ij}(t+1) = w_{ij}(t) + \eta \delta_i v_j$$



[This is w error being propagated to j unit]

$$w_{jk}(t+1) = w_{jk} + \eta \delta_j s_k \text{ (see above)}$$

$$\delta_j = v_j(1-v_j) \sum_{i=1}^n \delta_i w_{ij}$$



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